Chapter 4
The Stability Threshold

Let us recall the method of studying the asymptotic numerical stability of a linear system

$$\dot{z} = Az, \tag{4.1}$$

where $z \in \mathbb{R}^m, A \in \mathbb{R}^{m\times m}$, when solved by a numerical method. Typically one shows that under generic conditions analyzing the asymptotic behavior of the numerical method for solving the system can be related to analyzing scalar equations defined via an eigendecomposition of A, that is, for the purposes of stability analysis, we may replace the system (4.1) by m scalar equations of the form $\dot{u} = \lambda u$ where λ is an eigenvalue of A. Here λ, hence u may in general be complex. Then one applies the numerical method directly to the scalar equation. For example Euler's method yields

$$u_{n+1} = (1 + h\lambda)u_n.$$

The iteration is stable if $|u_n|$ remains bounded as $n \to \infty$, implying the following condition for asymptotic stability:

$$|1 + h\lambda| \leq 1.$$

This defines the "stability region" for Euler's method as a disk in the complex $h\lambda$-plane centered around -1 of radius 1. This region is shown in Fig. 4.1. If $h\lambda$, where λ is an eigenvalue of A lies in the indicated stability region, then Euler's method will be stable for the linear system $\dot{z} = Az$.

If we consider applying this technique to the harmonic oscillator $\dot{q} = p; \dot{p} = -\Omega^2 q$, then the matrix A is

$$A = \begin{bmatrix} 0 & I \\ -\Omega^2 & 0 \end{bmatrix}.$$

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